

Intelligent RF Spectrum Scan Strategy

A wideband receiver can listen to only 2 of 16 bands at a time. We learn which 2 — and when.

PROBLEM STATEMENT ID

[fill in from the portal]

THEME

[fill in from the portal]

PROBLEM STATEMENT TITLE

Intelligent RF Spectrum Scan Strategy

PS CATEGORY

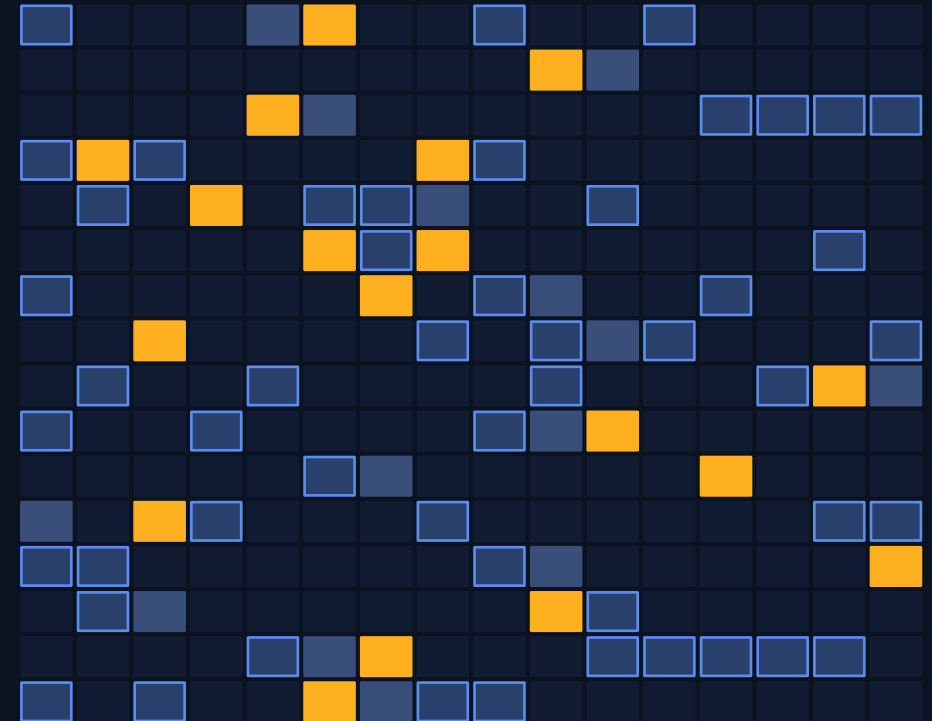
Software

TEAM ID

[fill in from the portal]

TEAM NAME

[your registered team name]



■ Heard — receiver tuned here, emitter active

■ Tuned, but the band was quiet

■ Active, and missed — nobody was listening

16 bands across · time flowing down · $K = 2$ tuned at any instant

SHRUTI — learning where to listen, and when

Today the receiver sweeps bands in a fixed order. **Four out of five dwells land on empty air**, and short bursts are lost between visits. SHRUTI replaces the sweep with a policy that decides, every step, which bands are worth hearing next.

HOW THE SOLUTION WORKS

1 Model the spectrum

16 bands, 10 emitters across five behaviour classes — fixed, periodic, agile, random, intermittent. Deterministic by seed.

2 Predict the rhythm

Ai-ml-2 folds each band's detection times onto candidate periods and returns a next-activation window with a calibrated confidence.

3 Decide where to listen

Ai-ml-1 reads per-band state — time since scan, detection EWMA, misses, periodicity phase — and picks the next K bands.

4 Learn from the outcome

Each step returns a reward and updates the policy online. Scoring uses ground truth the scheduler is never shown.

INNOVATION AND UNIQUENESS

■ We observe our detections, not the emitter.

The median gap between detections measures the scan schedule, not the signal. Phase folding recovers the true period even when whole cycles went unheard.

■ Confidence that survives the search.

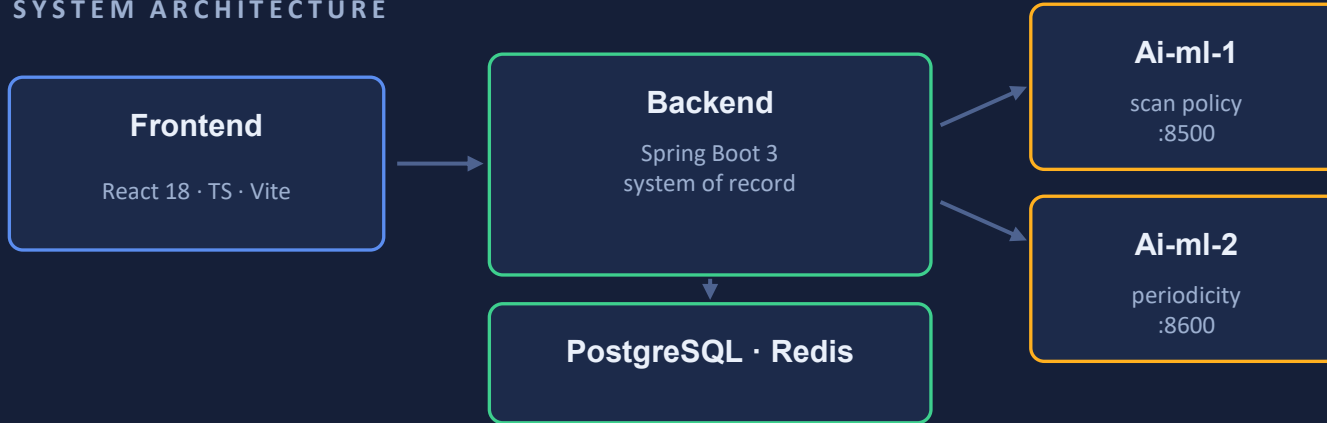
p-values are corrected for the number of independent trial periods, so the four non-periodic emitter classes sit at the nominal 5%, not 99%.

■ The trade-off is a dial, not an accident.

One reward weight moves the policy monotonically between hearing more spectrum and catching more distinct bursts. Retuned per mission.

Four services, one contract, a decision every 6.6 ms

SYSTEM ARCHITECTURE



The Frontend talks only to the Backend. The Backend is the only caller of either ML service, and assembles the state vector before it asks for a decision.

API_CONTRACT.md is copied byte-identical into all four folders — one source of truth, four independently buildable teams.

PER-STEP DECISION LOOP

- 01 Advance the spectrum one step
 - 02 Ai-ml-2 predicts each band's next window
 - 03 Backend builds the state vector
 - 04** Ai-ml-1 returns the next K bands
 - 05 Scan, detect, score against ground truth
 - 06 Reward goes back — the policy updates
- 6.6 ms** p95 decide over HTTP, 50 ms budget
- 3.0 ms** p95 periodicity, 64 bands

TECHNOLOGY STACK

Backend

Java 21 Spring Boot 3 PostgreSQL Flyway Redis WebSocket

Machine learning

Python 3.12 FastAPI Gymnasium NumPy / SciPy Stable-Baselines3

Frontend and ops

React 18 TypeScript Vite Docker Prometheus pytest / JUnit

It runs today, and we know where it loses

WHAT ALREADY RUNS

- Four services, one command each, Dockerised, booting on H2 or Postgres with no other setup.
- Deterministic by seed — the same seed gives every policy an identical spectrum, so comparisons are like for like.
- A one-button scripted demo: live waterfall, running metrics, then the baseline-versus-learned comparison.
- DQN and PPO are built and verified behind a flag; the bandit is the default because it needs no pre-training.

167

tests, scheduler

77

tests, estimator

96%

coverage, Ai-ml-2

87%

coverage, Ai-ml-1

Simulation-only by design. The policy consumes an abstract per-band state vector, so no code path touches a radio.

CHALLENGES, AND HOW WE HANDLE THEM

● The learned policy does not win everything.

It catches fewer distinct bursts than the sweep — 0.174 against 0.413.

- A real resource frontier, not a defect. With $K = 2$ of 16 you either spread thin and clip the start of many bursts, or concentrate and hear far more air. One reward weight moves along it monotonically, and is retuned per mission.

● Raw intercept time ranks the better policy worse.

It averages only the bursts a policy actually caught.

- We report censored intercept time beside it, charging every undetected burst the full episode, so the average is defined over all bursts rather than the easy ones.

● The periodicity service cannot help end to end yet.

It is told when a detection happened, never when a band was heard and found silent.

- Diagnosed, with a named fix: carry scanned-but-empty observations in the contract. The acceptance test stays in the suite, disabled with this reason, so the gap cannot quietly go stale.

Three times the detection rate, from the same receiver

3.2×

Detection rate

Pd 0.107 → 0.345

3.7×

High-priority detection

HPDR 0.101 → 0.375

2.9×

Listening efficiency

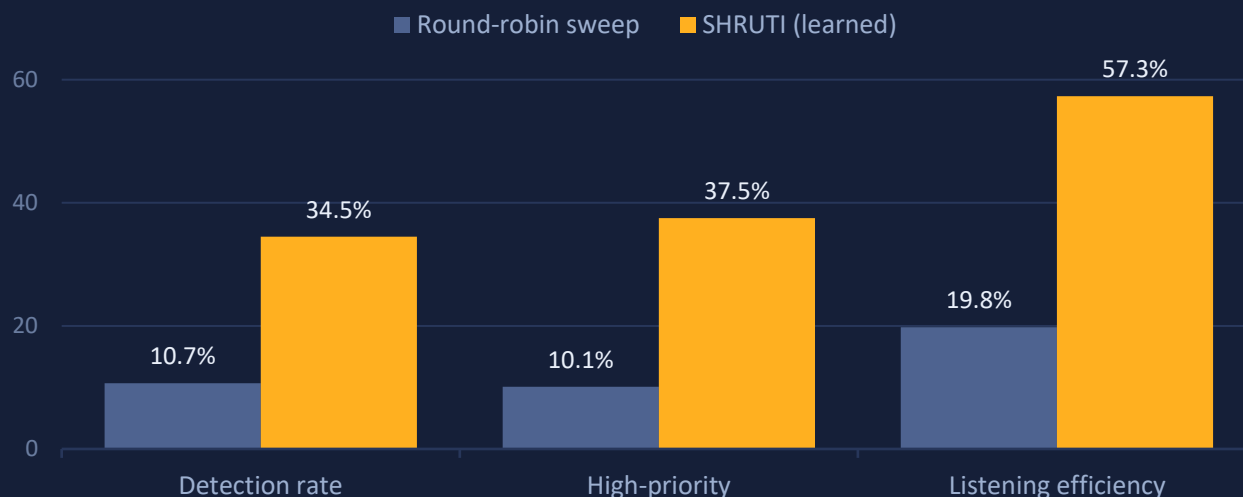
0.198 → 0.573 of dwell on live air

6.6 ms

Decision latency

p95 over HTTP, 50 ms budget

ROUND-ROBIN SWEEP VERSUS THE LEARNED POLICY



WHO IT SERVES

Operational

Spectrum-monitoring and EW operators hear more of the band plan from the receiver they already own — a capability gain in software, with no extra RF front-ends.

Economic

A reproducible testbed lets scan strategies be compared, and rejected, before any hardware is committed to a programme.

Scientific

The phase-folding estimator and the censored-average metric generalise to any sparse, irregularly sampled event series.

Measured against the baselines it replaces

SCENARIO B RESULTS — 5 SEEDS, 1000 STEPS

Metric	Random	Round-robin	SHRUTI
Detection rate (Pd)	0.104	0.107	0.345
High-priority detection	0.109	0.101	0.375
Listening efficiency	0.190	0.198	0.573
False-alarm rate (Pfa)	0.069	0.068	0.070
Distinct bursts caught	0.340	0.413	0.174

Green marks where the sweep still wins. We report it rather than dropping the row.

Prototype and source

Repository: [[add your GitHub link](#)] Demo video: [[add your link](#)]

scripts/compare.py reproduces every number on this slide from a clean checkout.

PRIOR ART

- US 10,523,342 B1 — Autonomous RL method of receiver scan schedule control. BAE Systems.
- Yücek and Arslan, A survey of spectrum sensing algorithms for cognitive radio. IEEE Comms. Surveys and Tutorials, 2009.
- Survey of wideband spectrum sensing and sub-Nyquist approaches. arXiv:2001.02574.

DECISION POLICY

- Sutton and Barto, Reinforcement Learning: An Introduction, 2nd ed. MIT Press, 2018.
- Auer, Cesa-Bianchi and Fischer, Finite-time analysis of the multiarmed bandit problem. Machine Learning 47, 2002.
- Mnih et al., Human-level control through deep reinforcement learning. Nature 518, 2015.

PERIOD ESTIMATION

- Mardia and Jupp, Directional Statistics. Wiley, 2000 — Rayleigh test, mean resultant length.
- VanderPlas, Understanding the Lomb-Scargle periodogram. ApJS 236, 2018.